

Lecture 2 review

Tutorial 2 — Sigma-algebras and measures

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A **sigma-algebra** $\mathcal{T} \subseteq \mathcal{P}(\Omega)$ satisfies

$$\Omega \in \mathcal{T}, \quad A \in \mathcal{T} \Rightarrow A^c \in \mathcal{T}, \quad (A_n)_{n \geq 1} \subseteq \mathcal{T} \Rightarrow \bigcup_{n \geq 1} A_n \in \mathcal{T}.$$

It also contains $\emptyset$ and is closed under countable intersections and set differences.

Two examples

$\{\emptyset, \Omega\}$ and $\mathcal{P}(\Omega)$ are sigma-algebras.

Generated sigma-algebras and Borel sets

For $\mathcal{G} \subseteq \mathcal{P}(\Omega)$, the **generated sigma-algebra** is

$$\sigma(\mathcal{G}) = \bigcap \{ \mathcal{T} : \mathcal{T} \text{ is a sigma-algebra on } \Omega, \mathcal{G} \subseteq \mathcal{T} \}.$$

It is the smallest sigma-algebra containing $\mathcal{G}$.

On $\mathbb{R}$

$$\mathcal{B}(\mathbb{R}) = \sigma(\text{open sets}).$$

Generators of the Borel sigma-algebra

Each of the following families generates $\mathcal{B}(\mathbb{R})$:

- open sets; closed sets;
- open intervals (a, b) ; closed intervals $[a, b]$, with $a < b$;
- half-lines $(-\infty, a)$; $(-\infty, a]$;
- half-lines $(a, +\infty)$; $[a, +\infty)$.

In these interval families, endpoints may range over all real numbers or only over rational numbers.

In $\mathbb{R}^n$

Open sets, balls with rational centres and radii, open boxes and half-open boxes with rational endpoints each generate $\mathcal{B}(\mathbb{R}^n)$, for example

$$\prod_{i=1}^n (a_i, b_i) \quad \text{or} \quad \prod_{i=1}^n (a_i, b_i], \quad a_i, b_i \in \mathbb{Q}, \quad a_i < b_i.$$

Definition of a measure

On a measurable space $(\Omega, \mathcal{T})$, a **measure** is a map $\mu : \mathcal{T} \rightarrow [0, +\infty]$ such that

$$\mu(\emptyset) = 0, \quad \mu\left(\bigsqcup_{n \geq 1} A_n\right) = \sum_{n \geq 1} \mu(A_n)$$

for every sequence of pairwise disjoint measurable sets.

Assumptions to check

Countable additivity applies to a *countable, pairwise disjoint* family. A measure may take the value $+\infty$.

Examples of measures

- **Dirac measure at x :** $\delta_x(A) = \mathbb{1}_A(x)$.
- **Counting measure:** $\mu(A) = \#A$ if A is finite, $+\infty$ otherwise.
- **Lebesgue measure:** $\lambda([a, b]) = b - a$ for $a \leq b$.

Elementary properties

For measurable sets,

$$\begin{aligned}A \subseteq B &\Rightarrow \mu(A) \leq \mu(B), \\ \mu\left(\bigcup_{n \geq 1} A_n\right) &\leq \sum_{n \geq 1} \mu(A_n), \\ \mu(A \cup B) + \mu(A \cap B) &= \mu(A) + \mu(B).\end{aligned}$$

If $A \subseteq B$ and $\mu(A) < +\infty$, then

$$\mu(B \setminus A) = \mu(B) - \mu(A).$$

Arithmetic with infinity

The expression $+\infty - (+\infty)$ is undefined.

Continuity of a measure

Increasing sequence

If $A_n \subseteq A_{n+1}$, then

$$\mu\left(\bigcup_{n \geq 1} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Decreasing sequence

If $A_{n+1} \subseteq A_n$ and $\mu(A_1) < +\infty$, then

$$\mu\left(\bigcap_{n \geq 1} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Without finiteness: $A_n = [n, +\infty)$ satisfies $A_n \downarrow \emptyset$, but $\lambda(A_n) = +\infty$.

Finite and sigma-finite measures

- μ is **finite** if $\mu(\Omega) < +\infty$.
- μ is **sigma-finite** if $\Omega = \bigcup_{n \geq 1} E_n$, where $E_n \in \mathcal{T}$ and $\mu(E_n) < +\infty$.

Examples

Lebesgue measure on $\mathbb{R}$ is sigma-finite: $\mathbb{R} = \bigcup_{n \geq 1} [-n, n]$.

Counting measure on $\mathbb{N}$ is sigma-finite. On an uncountable set with its power-set sigma-algebra, it is not.

A *measurable* set N with $\mu(N) = 0$ is a null set. For a measure that is not complete, a subset of a measurable null set need not be measurable.